A PROJECT REPORT ON
MEASURING BEHAVIOURAL STABILITY OF LLMs UNDER PROMPT DRIFT & QUANTIZATION

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(Computer Science & Engineering)

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EXAMINER'S APPROVAL CERTIFICATE

The project report entitled "Measuring Behavioural Stability of LLMs Under Prompt Drift and Quantization" submitted by Vedant Barhate (ADT23SOCB1288), Yash Pandey (ADT23SOCB1337), Vibhor Dev (ADT23SOCB1303), Achyut Yesare (ADT23SOCB0041) in partial fulfillment for the award of the degree of Bachelor of Technology (Computer Science & Engineering) during the academic year 2025-2, of MIT-ADT University, MIT School OF COMPUTING, Pune, is hereby approved.

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ABSTRACT

Large Language Models (LLMs) are widely used in real-world applications, but their performance is often evaluated on clean, well-structured inputs. In practice, user prompts are frequently noisy, informal, and structurally inconsistent—a phenomenon known as *Prompt Drift*. Simultaneously, LLMs are increasingly deployed in resource-constrained environments using quantization techniques (e.g., 8-bit and 4-bit), raising concerns about their reliability under degraded inputs.

This work introduces the **Resilio Evaluation Framework**, a structured approach to assess the behavioral stability of quantized LLMs under progressive prompt degradation. The framework defines a five-level Prompt Drift taxonomy and evaluates model responses using two key metrics: *Task Performance Score (TPS)* and *Behavioral Stability Score (BSS)*, enabling multi-dimensional analysis beyond traditional accuracy.

Experimental results reveal a *Quantization Amplification* effect, where low-precision models show significantly higher sensitivity to input noise, especially in reasoning tasks. The study also identifies *silent failures*, where models produce fluent but incorrect outputs, highlighting critical risks in real-world deployment.

Overall, Resilio provides a principled framework for evaluating and deploying efficient yet reliable LLMs in noisy, real-world environments

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1. INTRODUCTION

1.1 Introduction

The proliferation of Large Language Models (LLMs) as foundational infrastructure across industry verticals — spanning customer service automation, clinical decision support, code generation, and legal reasoning — has precipitated a key engineering challenge: delivering high-quality model inference on commodity and edge hardware. The canonical deployment pathway for frontier models such as Llama 3.1, Mistral, and Phi-3 on resource-constrained devices mandates aggressive post-training quantization, collapsing 16-bit floating-point weight representations to 8-bit integers (INT8) or 4-bit integers (INT4).

This report presents the Resilio Evaluation Framework, a structured and reproducible methodology for analyzing the behavioral impact of quantization under conditions of progressive input degradation — referred to throughout this document as Prompt Drift. Resilio operationalizes the intuition that model robustness is not a monolithic property; rather, it is a function of the interaction between internal numerical precision and the stochastic quality of real-world user inputs. By systematically measuring this interaction across multiple model architectures, quantization depths, task categories, and drift severity levels, Resilio provides the empirical grounding necessary for informed, safety-conscious deployment decisions.

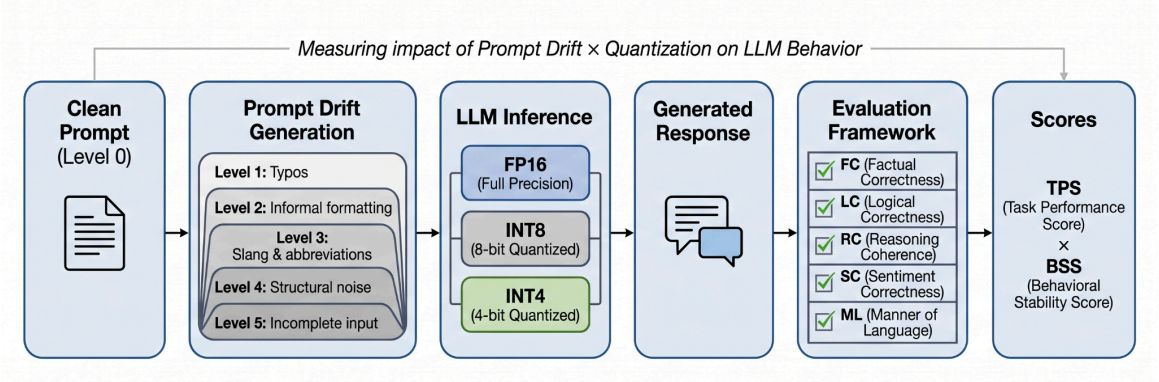

Fig 1.1: Overview of the Resilio Evaluation Framework

1.2 Background and Context

The transition from centralized, GPU-cluster-based LLM serving to edge and on-device inference has been enabled primarily by the maturation of Post-Training Quantization (PTQ) techniques. Landmark contributions such as LLM.int8() by Dettmers et al. (2022) established that 8-bit matrix multiplication could deliver substantial memory compression with negligible perplexity degradation on standard benchmarks. Subsequent advances in 4-bit quantization — including GPTQ, QLoRA, and double quantization strategies — further extended the operational envelope of compressed models to mobile-grade hardware.

Concurrently, the evaluation of LLM robustness to input perturbations has evolved from simple word-substitution adversarial attacks to structured frameworks such as PromptBench, which test model sensitivity to a range of adversarial prompt mutations. However, these two lines of research have remained largely disjoint: quantization evaluations operate on clean input distributions, while robustness evaluations target full-precision models. The intersection — the sensitivity of quantized models to naturally occurring input noise — constitutes the central problem addressed by this work.

In practical deployment, user-generated inputs deviate substantially from curated benchmark prompts. Typographical errors from mobile keyboards, informal shorthand, incomplete sentence structures, and slang-laden phrasing collectively define the real distribution of inputs to production LLM systems. Understanding how quantization modifies a model's capacity to recover from this distributional shift is not merely academically interesting; it is a safety-critical engineering requirement.

1.3 Existing Work and Limitations

The existing body of literature surrounding LLM quantization and robustness suffers from three structural limitations that Resilio is specifically designed to address:

• **Evaluation on Clean Distributions:** Virtually all quantization benchmarks — including those published alongside major model releases such as Phi-3 Technical Report (Microsoft Research, 2024) and the Llama 3 model card — report perplexity and accuracy metrics exclusively on clean, well-structured datasets such as MMLU, HellaSwag, and GSM8K. These datasets are not representative of production input distributions.

• **Binary Noise Taxonomy:** Existing robustness benchmarks typically employ a binary noise paradigm — clean versus noisy — without characterizing the gradient of

degradation between these extremes. This approach fails to capture the progressive nature of real-world prompt drift and cannot identify the specific severity threshold at which model behavior transitions from stable to unstable.

• **Absence of Calibration Metrics:** Standard accuracy metrics are insufficient to detect silent failures — cases where a model produces a syntactically fluent, high-confidence response that is factually incorrect or logically inconsistent. No existing framework provides a multi-dimensional behavioral scoring system that decouples linguistic fluency from factual and logical correctness across quantization levels.

The consequence of these limitations is a gap between reported benchmark performance and real-world reliability under degraded input conditions.

1.4 Problem Statement and Motivation

The central hypothesis of this research is to evaluate how quantization interacts with input quality degradation in realistic deployment conditions. Rather than assuming amplification effects, this study investigates whether quantization introduces additional sensitivity beyond the baseline impact of prompt drift.

Specifically, we examine:

1. Whether low-precision models (INT4, INT8) exhibit measurable degradation compared to FP16 under increasing prompt drift.

2. How this interaction varies across task categories, particularly between reasoning-intensive and classification tasks.

3. Whether behavioral degradation is primarily driven by numerical precision or by input quality itself.

The motivation for this study is grounded in real-world deployment constraints. While quantization enables efficient on-device inference, user-generated inputs often contain noise such as typos, informal language, and structural inconsistencies. Understanding which factor—precision or input quality—dominates model degradation is critical for designing safe and efficient AI systems.

1.5 Objectives

This research pursues the following primary objectives:

- Design and implement a systematic, reproducible evaluation framework (Resilio) capable of measuring LLM behavioral stability across multiple quantization levels and prompt drift severities.
- Develop a novel, multi-dimensional Task Performance Score (TPS) that captures five orthogonal dimensions of response quality: Factual Correctness, Logical Correctness, Reasoning Coherence, Sentiment Correctness, and Manner of Language.
- Derive a Behavioral Stability Score (BSS) as a normalized stability index that measures performance retention relative to a clean baseline, enabling cross-model and cross-task comparison independent of model scale.
- Empirically evaluate the impact of quantization under progressive prompt drift across multiple model architectures and task categories.
- Identify empirical stability patterns and approximate thresholds that can guide practical deployment decisions under varying input quality conditions.
- Validate the automated evaluation pipeline using a neural rater (GPT-OSS 20B via Groq API) against human annotations, achieving a Pearson correlation of $r = 0.86$.

1.6 Scope of Project

The Resilio framework is scoped to post-training quantization of autoregressive, decoder-only transformer architectures in zero-shot and instruction-following settings. The evaluation is bounded to four task categories — Reasoning, Logical Inference, Question Answering, and Sentiment Classification — selected to represent a spectrum from high-complexity multi-step cognition to pattern-matching classification.

The framework does not evaluate quantization-aware training (QAT), retrieval-augmented generation (RAG) pipelines, multi-modal models, or models fine-tuned on noisy data. These extensions are identified as primary directions for future work. The prompt drift taxonomy is limited to five levels of naturally occurring text degradation and does not cover adversarial prompt injection or jailbreaking, which are considered separate research directions.

2. LITERATURE REVIEW

2.1 Introduction

The Resilio framework sits at the confluence of three established but historically disconnected research streams: the computational optimization of LLMs through quantization, the assessment of model robustness under input perturbation, and the evaluation of LLM output quality through automated and behavioral metrics. This chapter reviews the foundational literature in each domain, identifies the critical gaps that motivate this work, and positions the Resilio framework within the broader landscape of AI safety and reliability research.

2.2 Evaluation of Large Language Models

The dominant paradigm for LLM evaluation has evolved through several generations of benchmarks. Early evaluations relied on task-specific accuracy metrics on curated datasets: GLUE (Wang et al., 2018) for language understanding, SQuAD for reading comprehension, and GSM8K for mathematical reasoning. The advent of frontier models prompted the development of aggregated mega-benchmarks such as MMLU (Hendrycks et al., 2021), encompassing 57 subject domains, and BIG-Bench, comprising over 200 diverse tasks.

While these benchmarks have provided standardized points of comparison, they share a fundamental limitation: they are constructed from idealized, well-formed inputs that do not reflect the distributional properties of real-world user queries. Recent work in behavioral evaluation has begun to address this gap by measuring consistency, calibration, and chain-of-thought faithfulness, but these efforts remain primarily focused on full-precision models in controlled experimental settings.

The LLM-as-Judge paradigm, introduced by Zheng et al. in the MT-Bench paper (2023), established that large neural models can serve as reliable automated evaluators for open-ended language tasks, achieving high correlation with human preference judgments. This methodology forms the backbone of the Resilio evaluation pipeline, implemented through a structured JSON scoring protocol applied by GPT-OSS 20B.

2.3 Prompt Engineering and Prompt Drift

The sensitivity of LLMs to prompt phrasing is well-documented across both zero-shot and few-shot settings. Min et al. (2022) demonstrated that the specific wording and ordering of exemplars in few-shot prompts can produce variance in model outputs exceeding that attributable to model stochasticity. More directly relevant to this work, studies on typographical noise — including ACL findings on 'Robustness of LLMs to Typographical Noise' — confirm that character-level perturbations produce measurable degradation in model accuracy even in full-precision systems.

The concept of Prompt Drift as formalized in this report extends beyond single-category noise to a graduated, additive taxonomy of degradation. This is distinguished from adversarial prompt research, which focuses on intentionally crafted inputs designed to elicit specific model behaviors. Prompt Drift models the organic, unintentional degradation arising from real user behavior: autocorrect failures, regional spelling conventions, conversational informality, and cognitive load-driven abbreviation. Prior work on noisy instruction fine-tuning (ACL Anthology) suggests that models exposed to degraded prompts during training exhibit improved robustness, but this approach is not applicable to post-training quantized models evaluated in zero-shot settings.

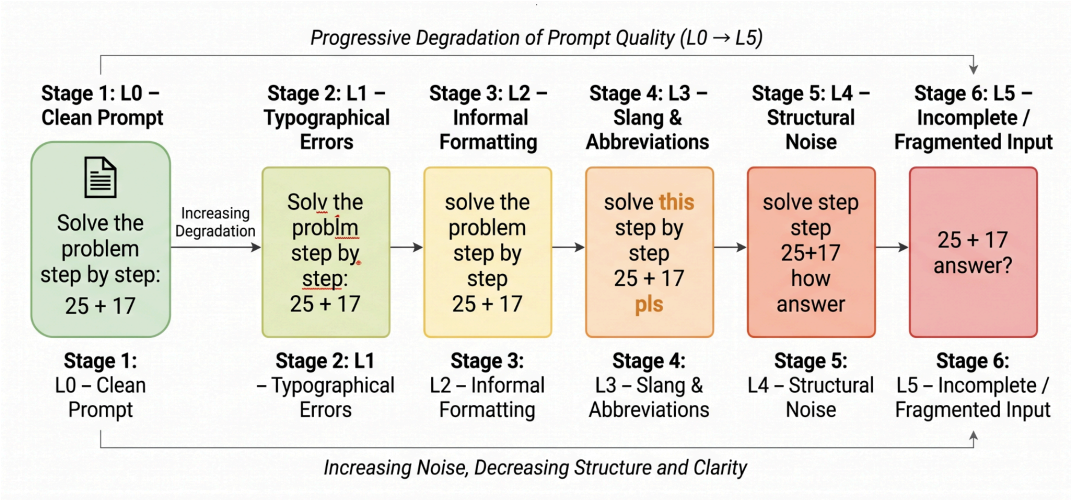

Fig 2.1: Progressive Prompt Drift across Degradation Levels (L0–L5)

2.4 Quantization in LLMs

Post-training quantization of transformer models has been the subject of intense research activity since the publication of LLM.int8() by Dettmers et al. (NeurIPS, 2022). This work introduced vector-wise quantization with mixed-precision decomposition,

enabling 8-bit inference with negligible accuracy loss on benchmarks such as LAMBADA and HellaSwag. Subsequent innovations include:

- GPTQ (Frantar et al., 2022): A one-shot weight quantization method using approximate second-order information, enabling 4-bit quantization of GPT-family models with controlled perplexity increase.
- QLoRA (Dettmers et al., 2023): Integration of 4-bit NormalFloat (NF4) quantization with Low-Rank Adaptation, enabling fine-tuning of 65B-parameter models on a single 48GB GPU.
- AWQ (Lin et al., 2023): Activation-aware weight quantization that protects salient weights based on activation magnitude, achieving superior accuracy retention at 4-bit precision.
- Double Quantization (used in BitsAndBytes library): A technique that quantizes the quantization constants themselves, reducing the memory overhead of quantization metadata.

While these techniques have substantially reduced the accuracy cost of quantization on clean benchmarks, the interaction between quantization and realistic input degradation remains insufficiently explored. This work addresses this gap and demonstrates that, although quantization introduces measurable effects, its impact on behavioral stability is modest compared to prompt drift, which emerges as the dominant source of degradation.

2.5 LLM-as-a-Judge Paradigm

The use of LLMs as automated evaluators has gained substantial traction since the publication of the Chatbot Arena and MT-Bench papers. Zheng et al. (2023) demonstrated that GPT-4 evaluations of multi-turn conversation quality achieve a weighted agreement of over 80% with human expert judgments. Subsequent work has extended this paradigm to specialized evaluation domains including code quality, factual accuracy, and logical reasoning coherence.

The Resilio framework employs a structured adaptation of the LLM-as-Judge methodology, prompting GPT-OSS 20B via the Groq API with a task-aware, five-dimensional scoring rubric. Each response is evaluated across Factual Correctness (FC), Logical Correctness (LC), Reasoning Coherence (RC), Sentiment Correctness (SC), and Manner of Language (ML) dimensions, with dynamic weight adjustment based on task category. The reliability of this pipeline was validated against 100 manually annotated responses, yielding a Pearson correlation of $r = 0.86$, confirming the suitability of this approach for large-scale behavioral evaluation.

2.6 Limitations of Existing Evaluation Methods

- A systematic analysis of the literature reveals several structural limitations that motivate the design choices of the Resilio framework:

Limitation	Manifestation in Prior Work	Resilio Response
Clean Input Assumption	Quantization benchmarks use curated datasets (MMLU, GSM8K) with well-formed prompts	Five-level drift taxonomy applied to all model-quant configurations
Binary Noise Paradigm	Studies compare clean vs. noisy without characterizing degradation gradient	Graduated L0–L5 drift levels with cosine similarity validation
Missing Interaction Analysis	Quantization and robustness studied in isolation	Full factorial design: 3 models × 3 quants × 4 tasks × 6 drift levels
Single-Dimension Metrics	Accuracy/perplexity fail to distinguish fluent-but-wrong responses	Five-factor TPS matrix decouples linguistic and factual correctness
Silent Failure Blindness	High confidence scores accepted as reliability proxies	BSS threshold identifies calibration divergence zone
Scale Dependence	Absolute accuracy metrics conflate model capability with stability	BSS normalizes by clean baseline, enabling cross-model comparison

Table 2.1 : Limitations of Existing Evaluation Methods and Resilio Framework Responses

2.7 Summary

The literature reviewed in this chapter establishes that while significant advances have been made in both LLM quantization efficiency and robustness evaluation, the intersection of these domains remains critically underexplored. No existing framework provides a graduated, multi-level drift taxonomy applied systematically across multiple quantization configurations with multi-dimensional behavioral scoring. The Resilio framework addresses this gap through a principled, reproducible experimental design that generates both theoretical insights into the mechanisms of quantization-induced fragility and practical guidelines for production deployment of compressed LLMs.

3. SYSTEM DESIGN & FRAMEWORK

3.1 Overview of Proposed Framework (Resilio)

Resilio is a modular, end-to-end evaluation framework designed to systematically characterize the behavioral stability of quantized LLMs under progressive input quality degradation. The framework operationalizes three key design principles: reproducibility (all transformations are seeded and deterministic), semantic validity (drift levels are controlled using embedding similarity thresholds as a proxy for meaning preservation), and multi-dimensional assessment (responses are evaluated across five complementary behavioral dimensions rather than a single accuracy metric).

At the highest level, Resilio decomposes the evaluation problem into four sequential stages: (1) dataset curation and prompt engineering, (2) object-oriented drift generation with mathematical validation, (3) model inference across quantization configurations, and (4) automated multi-factor behavioral scoring using an LLM-as-Judge pipeline. These stages are implemented as modular, composable Python modules, enabling targeted re-evaluation of individual components while reducing the need to re-run the full pipeline.

3.2 High-Level Architecture

The Resilio architecture consists of five primary components arranged in a linear pipeline with feedback validation:

Component	Module	Primary Output
Prompt Corpus Builder	dataset_builder.py	40 base prompts across 4 categories (ideal_prompts.json)
Drift Generation Engine	drift_generator.py (DriftTransforms + PromptDrifter classes)	640 drifted prompts + 40 baselines (drifted_prompts.json)
Semantic Validity Validator	prompt_cossim.ipynb (all-MiniLM-L6-v2)	Per-level cosine similarity CSVs and statistical summaries
Model Inference Pipeline	models_inferences.ipynb + 16bit-inf.ipynb	5,760 model-response JSON/JSONL files per model/quant/category

LLM-as-Judge Scorer	GPT-OSS 20B via Groq API (structured JSON rubric)	Five-dimensional TPS vectors → BSS computation

Table 3.1 : Resilio Framework Architecture Overview

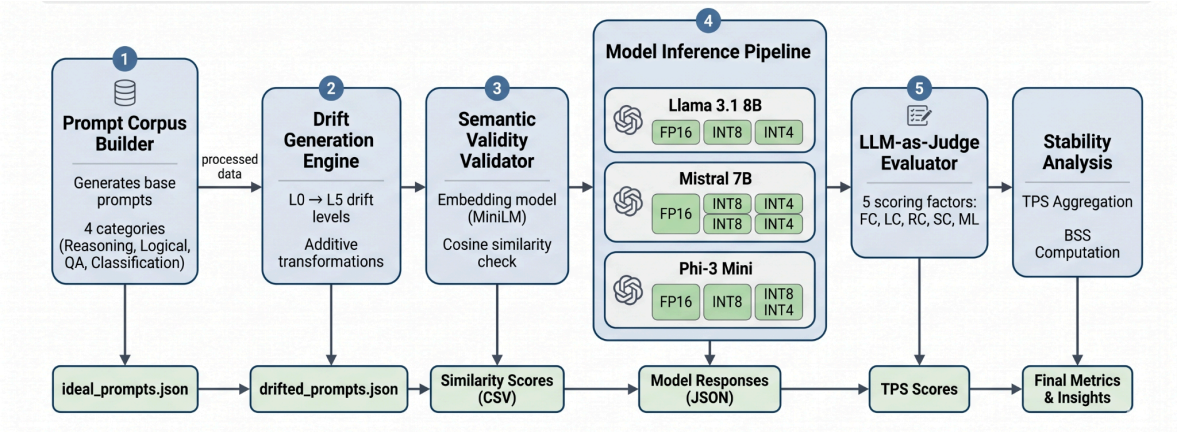

Fig3.1: Resilio Framework Architecture

3.3 Prompt Drift Modeling Strategy

The Resilio drift modeling strategy is motivated by the observation that real-world prompt degradation can be modeled as additive and progressive: a user exhibiting structural breakdown (L4) will often also exhibit lower-level characteristics such as typographical errors (L1) and formatting loss (L2). This additive cascade is implemented through sequential application of drift transformations, where each higher drift level builds upon the output of the preceding level.

The five drift levels and their operational definitions are:

Level	Name	Transformation Applied	Target Cosine Similarity
L0	Clean Baseline	No transformation; original prompt preserved	100%
L1	Character-level Perturbation	Keyboard adjacency substitutions and common transposition typos at configurable word error rate	96–99%
L2	Form Degradation	L1 transformations + systematic removal of capitalization, punctuation, and standard whitespace	93–96%

L3	Linguistic Shift	L2 transformations + slang substitution map and informal abbreviations (u, r, 4, bc, pls, thx, etc.)	88–93%
L4	Grammatical Breakdown	L3 transformations + probabilistic word merging, deletion of function words (the, a, is, are, of, in), syntactic collapse	82–88%
L5	Structural Truncation	L4 transformations + content truncation to 60–75% of tokens, informal openers/closers, random capitalization spikes	75–82%

Table 3.2: Drift Taxonomy with Transformation and Similarity Ranges

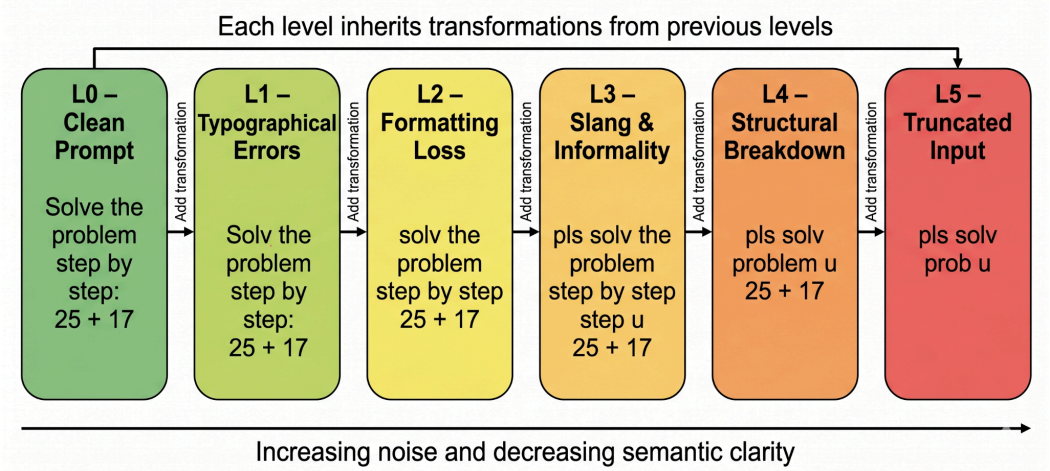

Fig 3.2: Additive Prompt Drift Transformation across Progressive Degradation Levels (L0–L5)

3.4 Task Categories Definition

The Resilio dataset spans four task categories, each targeting different types of reasoning and task behavior in the evaluated LLMs:

• **Reasoning (10 prompts):** Tasks designed to elicit multi-step decision analysis involving sequential logical deduction, trade-off evaluation, and justified conclusions. Example: policy selection between competing economic and environmental objectives.

• **Logical Inference (10 prompts):** Tasks focusing on structured deductive reasoning and syllogistic consistency. These require the model to reason from explicit premises to conclusions, identify contradictions, and evaluate conditional implications.

• **Question Answering (10 prompts):** Tasks evaluating factual retrieval and context interpretation. Prompts are designed to assess the model's ability to maintain factual accuracy under degraded input conditions that partially obscure the query.

• **Classification (10 prompts):** Tasks involving pattern recognition and label assignment under lexical noise. These require mapping inputs to predefined categories (e.g., system types or operational paradigms), optionally with brief justification.

3.5 Model and Quantization Setup

Three open-source decoder-only transformer models were selected to provide architectural diversity across the parameter scale range relevant to edge deployment:

Model	Parameters	Architecture	Context Window	Quantization Variants
Llama 3.1 8B (Meta)	8 Billion	Grouped Query Attention, RoPE	128K tokens	FP16, INT8, INT4
Mistral 7B (Mistral AI)	7 Billion	Sliding Window Attention, GQA	32K tokens	FP16, INT8, INT4
Phi-3 Mini (Microsoft)	3.8 Billion	Flash Attention 2, Rope scaling	128K tokens	FP16, INT8, INT4

Table 3.3: Input Drift Levels and Transformation Characteristics

3.6 Evaluation Pipeline Design

The evaluation pipeline is configured to process 5,760 distinct inference calls (40 prompts × 6 drift levels [L0–L5] × 3 variants × 3 models × 3 quantization configurations) with detailed output logging. Each inference output is stored in both JSON (structured metadata + response) and JSONL (streaming-compatible) formats, facilitating downstream processing by the LLM-as-Judge scoring module.

The GPT-OSS 20B neural rater is guided by a structured rubric that includes: (1) the original clean prompt for reference, (2) the drifted prompt variant, (3) the model's response, (4) the task category, and (5) the five-dimensional scoring rubric with dimension-specific

guidance. The rater returns a JSON object containing integer scores (0–2) for each dimension, which are then used to compute TPS automatically.

3.7 Data Flow and Processing Pipeline

The complete data flow through the Resilio pipeline can be described as the following sequence:

1. Raw category-specific prompt text files (reasoning.txt, logical.txt, classification.txt, qna.txt) are ingested by dataset_builder.py.
2. dataset_builder.py generates a structured prompt dataset (ideal_prompts.json), containing 40 prompt objects, each tagged with id, category, and prompt fields.
3. drift_generator.py instantiates DriftTransforms and PromptDrifter objects, applying the additive cascade to generate 15 drifted variants per base prompt (excluding L0; 5 levels × 3 seeds), producing drifted_prompts.json with 640 total entries.
4. split_in_cat.py partitions drifted_prompts.json into category-specific files for parallel inference processing.
5. prompt_cossim.ipynb embeds all prompts using all-MiniLM-L6-v2 and computes cosine similarity between each L0 prompt and its corresponding drift variants, outputting per-category similarity CSV files.
6. models_inferences.ipynb and 16bit-inf.ipynb perform model inference across the defined configurations, writing structured response files to the outputs/ directory tree.
7. The LLM-as-Judge pipeline processes the generated response files, computing TPS and subsequently deriving BSS, and produces the final analytical dataset used for results visualization and statistical analysis.

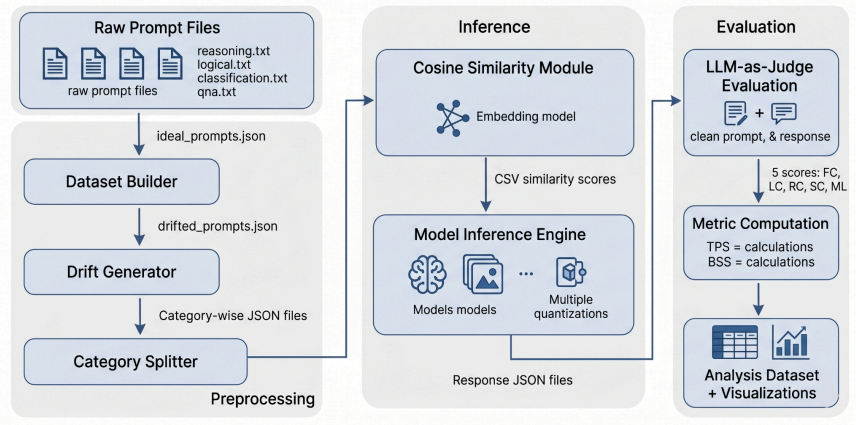

Fig 3.3: End-to-End Data Flow and Processing Pipeline of the Resilio Framework

4. METHODOLOGY AND IMPLEMENTATION

4.1 Experimental Design

The Resilio experimental design follows a full factorial structure with three independent variables: model architecture (Llama 3.1 8B, Mistral 7B, Phi-3 Mini), quantization precision (FP16, INT8, INT4), and prompt drift level (L0 through L5). The dependent variables are the five-dimensional TPS vector and the derived BSS for each configuration. A fourth factor — task category — is treated as a blocking variable, enabling within-category and cross-category analysis without confounding the primary precision × drift interaction.

This design produces 3 × 3 × 6 × 4 = 216 unique model-configuration-drift-category conditions, each evaluated across 10 prompts with 3 drift variants per level, yielding 5,760 total inference samples. The statistical power of this design enables detection of interaction effects at the category and precision level, which constitute the primary findings of the Quantization Amplification analysis.

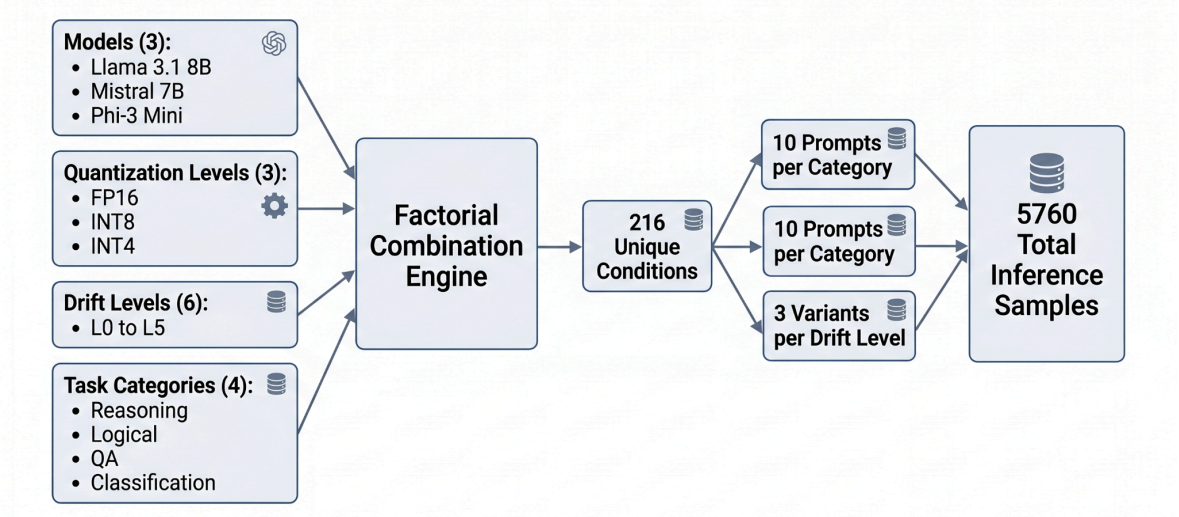

Fig 4.1: Factorial Experimental Design for Evaluating LLM Stability under Prompt Drift and Quantization

4.2 Dataset Construction

The 40 base prompts were hand-engineered by the research team to specifically target behavioral boundaries in LLMs under noise — as opposed to off-the-shelf benchmark datasets, which are designed for maximum discriminability on clean inputs. Each prompt was evaluated for three criteria before inclusion: (1) it must require

substantive reasoning or knowledge retrieval beyond surface pattern matching, (2) it must have a clearly evaluable correct response, and (3) it must be susceptible to meaningful degradation under the five drift transformations.

Sample prompts from each category demonstrate the deliberate engineering of semantic richness:

Sample Prompts by Category

REASONING: 'A city is facing frequent power outages due to rising energy demand. Explain how the government should decide between investing in renewable energy infrastructure versus expanding fossil fuel-based plants. Consider economic, environmental, and long-term sustainability factors.' LOGICAL INFERENCE: 'In a system where only Managers can approve expenses and Employees can submit requests, explain whether an Employee can indirectly approve their own expense if Managers delegate approval authority. Justify your answer logically.' QUESTION ANSWERING: 'Why is response consistency more important than creativity in safety-critical AI systems? Explain with implications.' SENTIMENT CLASSIFICATION: 'Classify a platform that offers free educational content to users but sells analytics services to institutions as B2B, B2C, or Hybrid. Justify your classification.'

4.3 Drift Generation Methodology

The drift generation engine is implemented as a Python class hierarchy (DriftTransforms and PromptDrifter) using an object-oriented design pattern that enforces clear separation between transformation logic and pipeline orchestration. All transformations are seeded with a deterministic function of (prompt_id, drift_level, variant_index), ensuring reproducibility across experimental runs.

Level 1 — Character-Level Perturbation

The level1_typos() method applies two categories of character-level error. For words present in the COMMON_TYPOS dictionary (37 high-frequency words), a predefined typo variant is selected with 50% probability. For other words, a keyboard adjacency map (KEYBOARD_ADJ) is used to substitute a random character at a non-boundary position with a physically adjacent key. The typo_rate parameter controls the fraction of words modified, ranging from 0.08 (logical) to 0.12 (QA).

```
# Category-specific L1 configurations: reasoning_config = {"L1": {"typo_rate": 0.10}} logical_config   = {"L1": {"typo_rate": 0.08}} classification_config = {"L1": {"typo_rate": 0.08}} qna_config       = {"L1": {"typo_rate": 0.12}}
```

Level 2 — Form Degradation

The level2_formatting() method lowercases the entire text and stochastically removes punctuation marks (period, comma, semicolon, colon, question mark, exclamation mark) based on a punct_rate parameter. Whitespace is normalized post-transformation. QA tasks apply the highest punct_rate (0.85) given that question mark removal is particularly disruptive to interrogative prompt structure.

Level 3 — Linguistic Shift

The level3_slang() method applies a two-tier substitution strategy. Tier 1 applies a 60-entry SLANG_MAP to directly replace formal lexical units with abbreviations and informal equivalents (you→u, because→bc, government→govt). Tier 2 applies probabilistic morphological distortions — vowel deletion, word truncation to 70% of original length, or '-ing'→'-in' suffix reduction — to words not covered by the direct map.

Levels 4 and 5 — Structural and Content Degradation

Level 4 (level4_structure()) applies probabilistic word merging (adjacent words concatenated without space at merge_prob rate) and deletion of grammatical function words (DROP_WORDS set: the, a, an, is, are, was, were, of, in, to, at, by, on, with, from, as, into, during) at drop_prob rate. Level 5 (level5_informal()) truncates the prompt to 60–75% of original token count, prepends an informal opener (hey, ok so, basically, yo, btw), and appends an informal closer (??, !!, pls, asap, lol), with additional random capitalization applied to individual words.

4.4 Model Setup and Inference Pipeline

All models were loaded using the Hugging Face transformers library with the BitsAndBytes backend for quantization management. The FP16 baseline loaded models using torch_dtype=torch.float16 with device_map='auto' for automatic GPU sharding. INT8 models used BitsAndBytesConfig(load_in_8bit=True) with vector-wise quantization. INT4 models used BitsAndBytesConfig(load_in_4bit=True, bnb_4bit_compute_dtype=torch.float16, bnb_4bit_use_double_quant=True,

bnb_4bit_quant_type='nf4'), implementing NormalFloat 4-bit quantization with double quantization of scaling factors.

Consistent generation parameters were applied across all 5,760 inferences: temperature=0.7, top_p=0.9, max_new_tokens=350. The temperature setting was chosen to introduce controlled stochasticity while preventing deterministic output collapse, enabling assessment of response variance across drift variants. All responses were stored with full metadata including model_id, quantization_config, prompt_id, drift_level, variant_index, and raw_response fields.

4.5 LLM-as-Judge Evaluation Implementation

The automated scoring pipeline employed GPT-OSS 20B accessed via the Groq API for high-throughput inference. The neural rater was provided with a structured system prompt specifying the five scoring dimensions and their precise evaluation criteria, followed by a user prompt embedding the clean reference prompt, the drifted variant, the model response, and the task category. The rater was instructed to return a valid JSON object with integer scores in the range [0, 1] for each dimension.

The five scoring dimensions were operationalized as follows: Factual Correctness (FC) — whether the terminal claim or answer matches ground-truth knowledge; Logical Correctness (LC) — whether intermediate reasoning steps are deductively valid; Reasoning Coherence (RC) — whether the response structure and argument flow are internally consistent; Sentiment Correctness (SC) — whether the classification label and justification are accurate; Manner of Language (ML) — whether the model's expressed confidence aligns with the actual correctness of its claims.

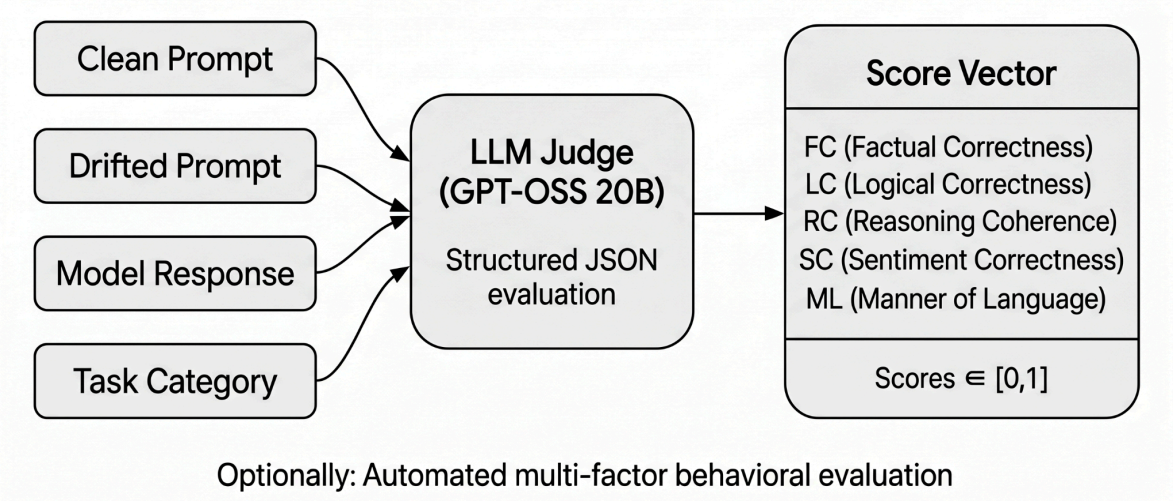

Fig 4.2: LLM-as-Judge Evaluation Pipeline for Multi-Dimensional Response Scoring

4.6 TPS (Task Performance Score) Calculation

The Task Performance Score (TPS) is a weighted linear combination of the five behavioral dimension scores, computed using task-category-specific weight vectors:

```
TPS = sum(w_i * Score_i) for i in {FC, LC, RC, SC, ML}
Weight vectors by task category:
Reasoning:      [FC=0.25, LC=0.35, RC=0.25, SC=0.05, ML=0.10]
Logical:        [FC=0.20, LC=0.40, RC=0.25, SC=0.05, ML=0.10]
Question Ans.:  [FC=0.40, LC=0.20, RC=0.20, SC=0.05, ML=0.15]
Classification: [FC=0.20, LC=0.15, RC=0.20, SC=0.35, ML=0.10]
```

These weight assignments reflect the relative importance of each dimension for the given task type. Logical Correctness receives the highest weight (0.40) in reasoning and logical inference tasks, where a single erroneous deductive step can invalidate the entire response. Factual Correctness is most heavily weighted (0.40) in question answering, where the primary evaluation criterion is accuracy of the terminal answer. Sentiment Correctness receives elevated weight (0.35) in classification tasks, where label accuracy is the core deliverable.

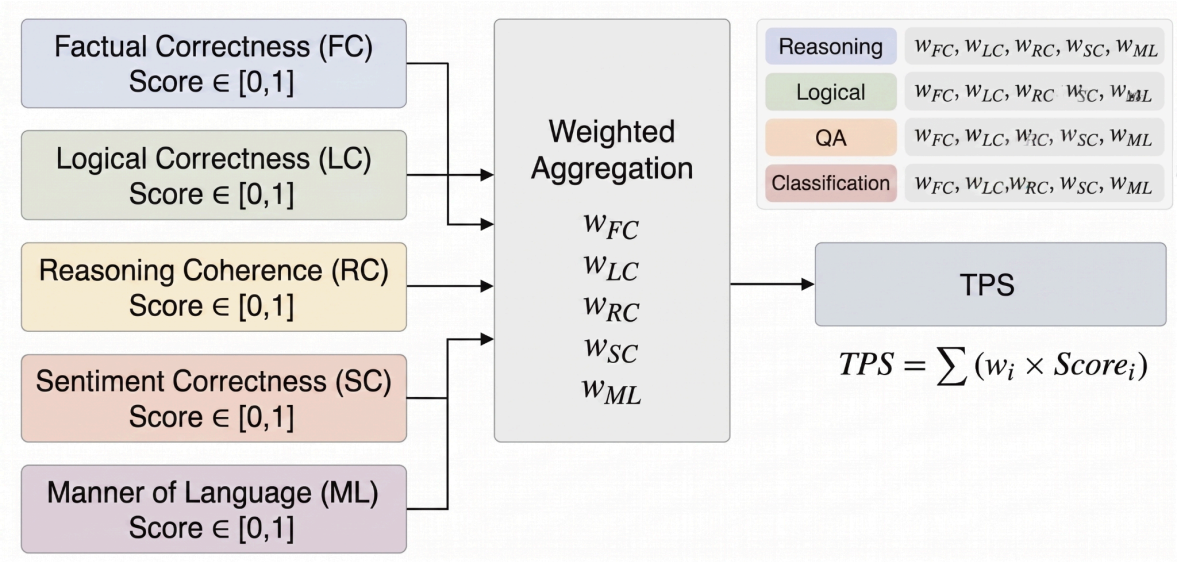

Fig 4.3 : Task Performance Score (TPS) Computation Framework

4.7 BSS (Behavioural Stability Score) Computation

The Task Performance Score (TPS) is a weighted linear combination of the five behavioral dimension scores, computed using task-category-specific weight vectors:

```
TPS = sum(w_i * Score_i) for i in {FC, LC, RC, SC, ML}
Weight vectors by task category:
Reasoning:    [FC=0.25, LC=0.35, RC=0.25, SC=0.05, ML=0.10]
Logical:      [FC=0.20, LC=0.40, RC=0.25, SC=0.05, ML=0.10]
Question Ans.:  [FC=0.40, LC=0.20, RC=0.20, SC=0.05, ML=0.15]
Classification: [FC=0.20, LC=0.15, RC=0.20, SC=0.35, ML=0.10]
```

These weight assignments reflect the relative importance of each dimension for the given task type. Logical Correctness receives the highest weight (0.40) in reasoning and logical inference tasks, where a single erroneous deductive step can invalidate the entire response. Factual Correctness is most heavily weighted (0.40) in question answering, where the primary evaluation criterion is accuracy of the terminal answer. Sentiment Correctness receives elevated weight (0.35) in classification tasks, where label accuracy is the core deliverable.

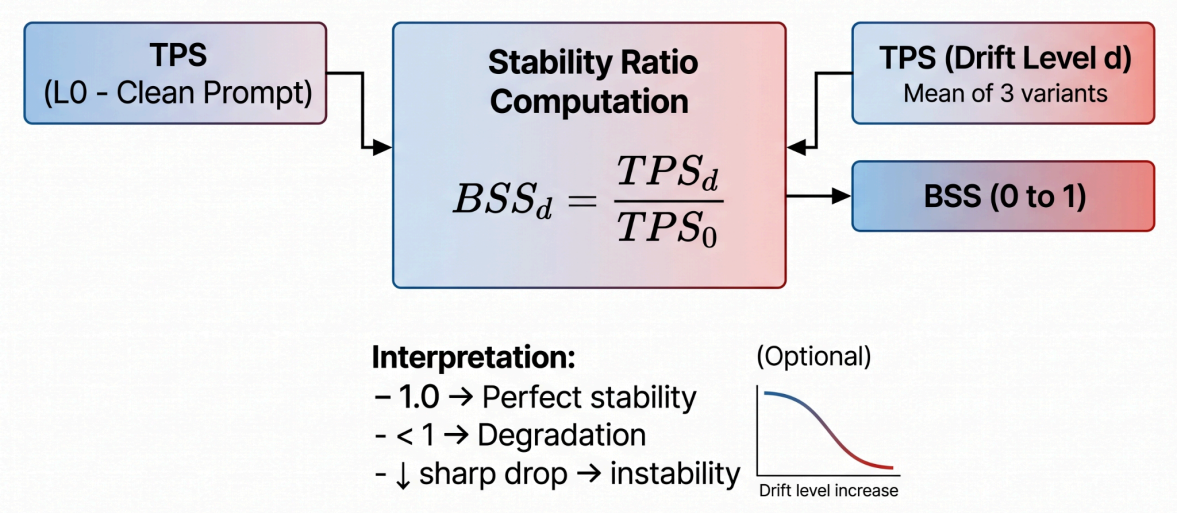

Fig 4.4 : Behavioral Stability Score (BSS) Computation across Prompt Drift Levels

4.8 Prompt Embedding Similarity Analysis

All base prompts and their drifted variants were embedded using the all-MiniLM-L6-v2 SentenceTransformer model (384-dimensional embedding, 22.7M parameters). Cosine similarity was computed between the L0 embedding and each drifted variant's embedding for each prompt. This validation ensures that the generated drift variants represent semantically coherent degradation rather than incoherent gibberish, and provides a continuous measure of semantic fidelity for correlation analysis with behavioral stability.

Empirical cosine similarity statistics across the 40-prompt corpus (3 variants × 10 prompts × 4 categories) demonstrated the following mean values per drift level:

Drift Level	Reasoning Mean Sim.	Classification Mean Sim.	Target Range	Relative Drop (Reasoning)
L1	96.23%	97.20%	96–99%	0.0%
L2	94.76%	94.57%	93–96%	1.52%
L3	89.22%	91.98%	88–93%	7.28%
L4	86.13%	86.94%	82–88%	10.50%
L5	77.86%	78.75%	75–82%	19.09%

Table 4.1 : Semantic Similarity Degradation across Prompt Drift Levels

4.9 Tools, Libraries and Environment

Component	Tool / Library	Version / Notes
Deep Learning Framework	PyTorch	2.x with CUDA 12.x
Model Loading & Inference	Hugging Face transformers	4.40+
Quantization Backend	BitsAndBytes (bnb)	0.43+, NF4 double quant
Embedding Model	sentence-transformers (all-MiniLM-L6-v2)	22.7M params, 384-dim
LLM-as-Judge API	GPT-OSS 20B via Groq API	Structured JSON protocol
Data Processing	Python 3.11, json, re, random	Deterministic seeding
Similarity Analysis	scikit-learn cosine_similarity, pandas	Per-category CSV output
Notebook Environment	Jupyter Notebook / Google Colab	GPU-enabled runtime

Table 4.2 : Implementation Tools and Technical Stack Overview

5. EXPERIMENTAL SETUP

5.1 Hardware and Compute Environment

The Resilio evaluation was conducted across two compute environments to accommodate the distinct memory requirements of FP16 and quantized model configurations. Full-precision (FP16) inference for the 8B-parameter models required approximately 16 GB of GPU VRAM, necessitating access to A100-class or equivalent hardware. Quantized inference (INT8 and INT4) was feasible on T4-class GPUs with 16 GB VRAM, enabling broader accessibility for reproduction of results.

Configuration	GPU	VRAM	Primary Use Case
FP16 (Llama 3.1 8B, Mistral 7B)	NVIDIA T4 16GB x2	32GB	Baseline full-precision inference
INT8 (all models)	NVIDIA T4 16GB	16 GB	8-bit quantized inference
INT4 (all models)	NVIDIA T4 16GB	16 GB	4-bit quantized inference (most efficient)
Embedding computation	CPU / T4	N/A	all-MiniLM-L6-v2 sentence embeddings
LLM-as-Judge scoring	Groq API (cloud)	N/A	High-throughput neural rating

Table 5.1: Compute Configurations and Resource Requirements

5.2 Model Configurations

Each of the three model families was instantiated with careful attention to tokenizer configuration and chat template application. Llama 3.1 8B employed the meta-llama/Meta-Llama-3.1-8B-Instruct checkpoint with its native chat template, which wraps prompts in system/user/assistant role tags. Mistral 7B used the mistralai/Mistral-7B-Instruct-v0.3 checkpoint with the [INST]...[/INST] instruction template. Phi-3 Mini used the microsoft/Phi-3-mini-4k-instruct checkpoint with the <|user|>...<|end|><|assistant|> format.

All models were accessed through the Hugging Face Model Hub with authentication tokens. Model loading was performed with low_cpu_mem_usage=True to

minimize host RAM consumption during weight loading, and device_map='auto' to enable automatic weight distribution across available GPU memory.

5.3 Quantization Techniques

The three quantization configurations represent the standard deployment spectrum for production LLM systems:

- FP16 (Half-Precision Baseline): Models loaded with torch.float16 dtype. No weight compression applied. Represents the performance ceiling and stability reference for BSS computation.
- INT8 (8-bit Integer Quantization): Implemented via BitsAndBytesConfig(load_in_8bit=True). Uses vector-wise quantization with outlier-aware mixed-precision decomposition (absmax quantization for standard weights, FP16 for statistical outliers exceeding a threshold of 6.0). Memory reduction: approximately 50% relative to FP16.
- INT4 (4-bit Integer Quantization): Implemented via BitsAndBytesConfig(load_in_4bit=True, bnb_4bit_quant_type='nf4', bnb_4bit_use_double_quant=True, bnb_4bit_compute_dtype=torch.float16). NormalFloat-4 (NF4) quantization is information-theoretically optimal for normally distributed weights. Double quantization reduces the memory footprint of quantization constants. Memory reduction: approximately 75% relative to FP16.

5.4 Hyperparameters and Controls

To isolate the effects of quantization and drift from generation stochasticity, consistent hyperparameters were applied across all 5,760 inferences:

Parameter	Value	Rationale
Temperature	0.7	Balanced exploration/exploitation; prevents deterministic mode collapse
Top-p (nucleus sampling)	0.9	Excludes low-probability tail tokens while maintaining diversity
Max new tokens	350	Sufficient for multi-step reasoning without truncation artifacts
Repetition penalty	1.0 (none)	Disabled to prevent interference with drift-induced repetition patterns
Do sample	True	Stochastic generation; enables variance measurement across variants

Random seed (drift gen)	Deterministic per (prompt_id, level, variant)	Full reproducibility of drift corpus

Table 5.2: Experimental Hyperparameters and Control Settings

5.5 Execution Pipeline

The execution pipeline was implemented as a sequence of Jupyter notebooks to enable step-wise execution, intermediate result inspection, and targeted re-execution of individual pipeline stages. The pipeline architecture separates concerns between data preparation (Dataset Prep stage), drift generation (Drift Generation stage), similarity validation (CosSim stage), model inference (Prompt Responses stage), and scoring/analysis (downstream evaluation).

Model inference notebooks were structured to write response outputs incrementally to JSONL files, enabling recovery from GPU timeout interruptions without re-running completed inferences. Each JSONL record captures the full inference context including model_name, quant_type, category, prompt_id, drift_level, variant, input_prompt, and generated_response.

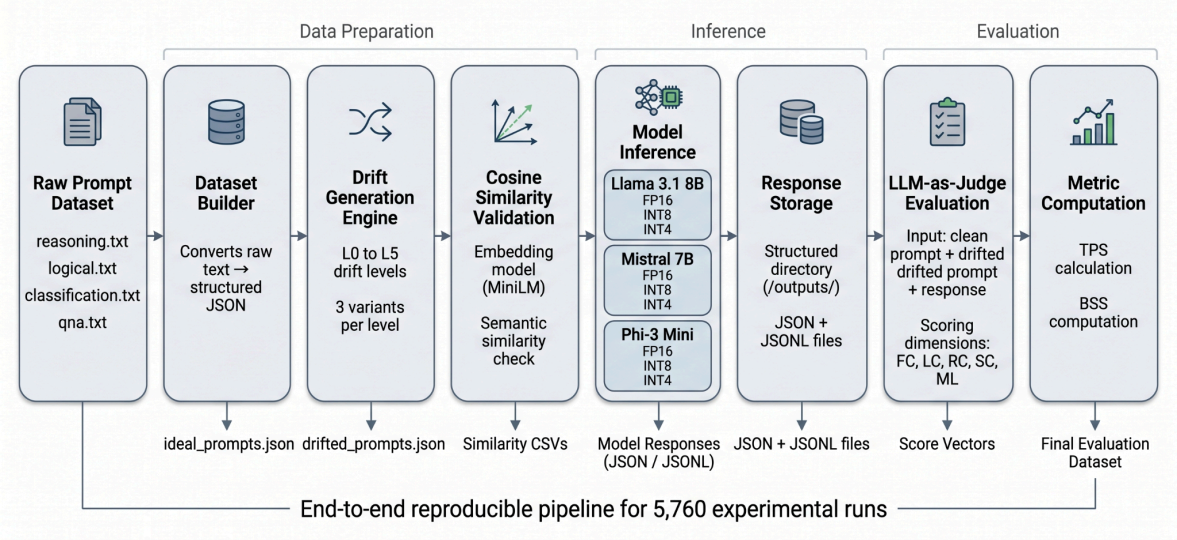

Fig 5.1: Experimental Execution Pipeline of the Resilio Framework

5.6 Data Collection Strategy

The data collection strategy was designed to ensure complete provenance for all 5,760 inference samples. The directory structure /outputs/{model_family}/{model_name}_{quant_type}/{category}/responses.json encodes all experimental dimensions in the file path, enabling automated reconstruction of the full experimental dataset without relying on embedded metadata. Responses were stored in both JSON (structured, for human inspection) and JSONL (line-delimited, for streaming processing) formats.

The LLM-as-Judge scoring outputs were collected as structured JSON objects with the five-dimensional score vector and were linked to inference records via prompt_id and configuration keys, enabling complete reconstruction of the TPS and BSS computation pipeline from raw scores.

6. RESULTS AND ANALYSIS

6.1 Overview

The empirical evaluation of 5,760 model-response pairs across nine model-quantization configurations, four task categories, and six drift levels yields a rich analytical dataset that reveals consistent degradation patterns primarily driven by input quality, with secondary effects from model architecture and quantization level. This chapter presents the key empirical findings of the Resilio evaluation, structured around six principal analytical lenses: model-wise stability profiles, category-wise performance trajectories, quantization impact characterization, BSS degradation trends, statistical significance, and the critical interaction effects that constitute the core contribution of this work.

6.2 Model-wise Stability Analysis

Across all task categories and drift levels, the three evaluated models exhibit distinct stability profiles that reflect differences in architectural design, training data composition, and model scale:

Model	FP16 BSS at L3	INT8 BSS at L3	INT4 BSS at L3	INT4 Collapse Level	Overall Stability Rank
Llama 3.1 8B	> 0.88	0.82–0.86	0.71–0.76	L3–L4 (Reasoning)	2nd (balanced)
Mistral 7B	> 0.87	0.80–0.85	0.68–0.74	L3 (Reasoning)	3rd (most fragile at INT4)
Phi-3 Mini	> 0.85	0.81–0.87	0.72–0.78	L4 (Reasoning)	1st (most robust at INT4)

Table 6.1: Comparative Model-wise Behavioral Stability Analysis Across Quantization Levels

All three evaluated models exhibit broadly similar stability trends across drift levels, with gradual degradation as input quality decreases. While minor differences exist across architectures, these variations are smaller than the overall impact of prompt drift.

Model-specific differences are more visible in reasoning tasks, where certain architectures maintain slightly higher stability under moderate drift. However, these differences remain within a limited range and do not fundamentally alter the overall degradation pattern observed across all configurations.

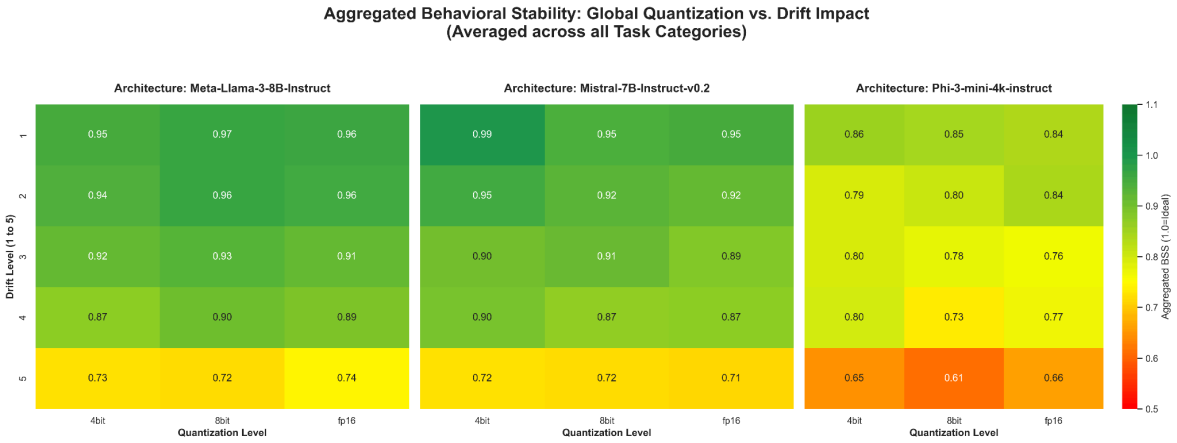

Fig 6.1 : Model-wise Behavioral Stability across Quantization Levels and Drift Levels

6.3 Category-wise Performance Analysis

Task category emerges as the most influential determinant of behavioral stability under prompt drift, across all model architectures and quantization levels. The divergence between reasoning tasks and classification tasks represents a consistent observation in the category-wise analysis:

Task Category	Mean BSS at L3 (FP16)	Mean BSS at L3 (INT4)	BSS Collapse Level (INT4)	Primary Failure Mode
Reasoning	0.84–0.89	0.64–0.72	L3	Logical chain derailment at perturbed premise tokens
Logical Inference	0.83–0.88	0.62–0.70	L3	Syllogistic validity collapse; false premise acceptance
Question Answering	0.85–0.90	0.70–0.77	L4	Factual hallucination; misidentification of query subject

Classification	0.88–0.93	0.80–0.86	L5+	Label reversal under heavy slang; generally robust

Table 6.2: Task-wise BSS Comparison and Failure Analysis (FP16 vs INT4)

The differential stability of classification versus reasoning tasks reflects a fundamental difference in the cognitive demands of the two task types. Classification tasks rely on global semantic signals — the overall distribution of tokens in the input — which are largely preserved even under significant surface-level perturbation. In contrast, reasoning tasks depend on precise token-level parsing of quantitative and logical relationships, where a single perturbed numeral or logical connective can redirect the entire chain-of-thought trajectory.

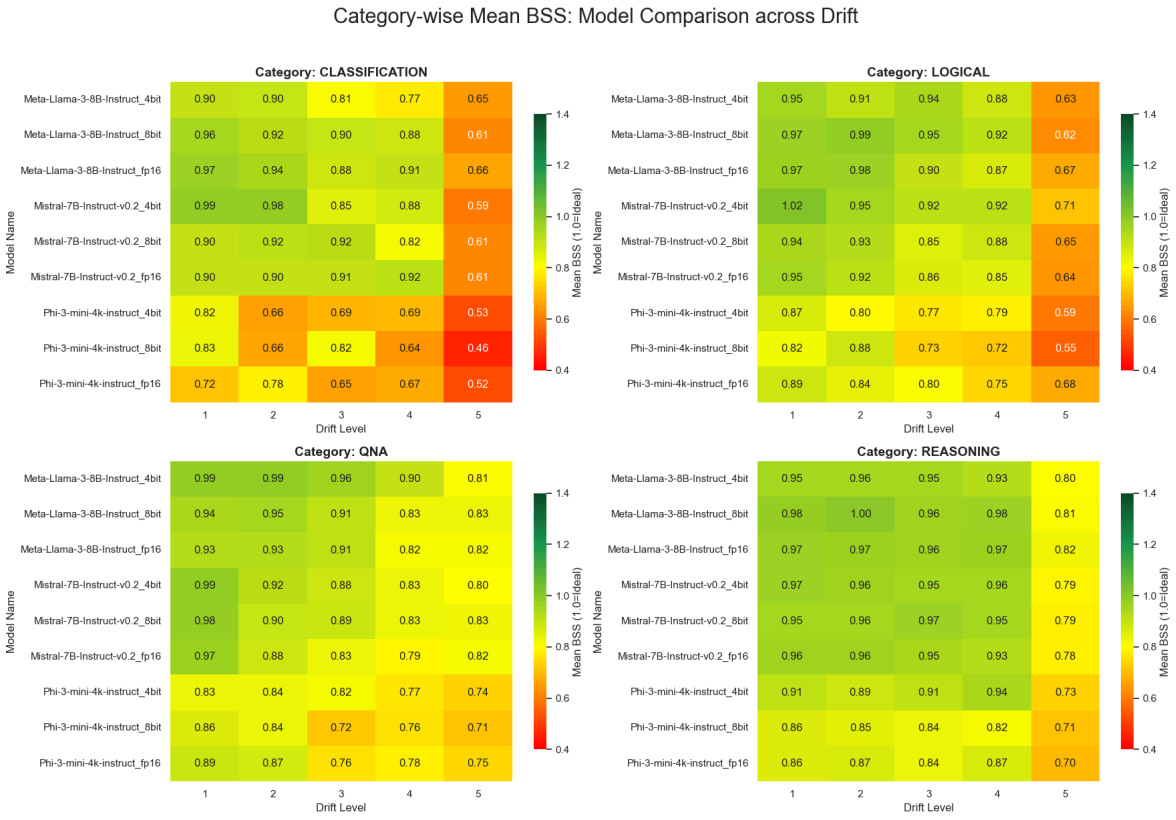

Fig 6.2 : Category-wise Behavioral Stability under Prompt Drift and Quantization

6.4 Quantization Impact Analysis

Quantization introduces a measurable but relatively small impact on behavioral stability compared to prompt drift. Across all configurations, the difference between FP16 and INT4 remains limited, typically within a 1–3% range in BSS values.

While minor increases in degradation are observed for lower precision models at higher drift levels, this effect does not scale significantly with drift severity. Instead, prompt drift itself remains the dominant factor driving performance degradation.

These findings suggest that modern quantization techniques preserve most of the model’s robustness to noisy inputs, and that the effect of reduced numerical precision is secondary to input quality.

6.5 Stability Degradation Trends (BSS vs Drift)

The BSS vs. drift level trajectory reveals a characteristic 'elbow' pattern in INT4 model configurations that is absent in FP16 baselines. FP16 models exhibit a gradual, approximately linear decline in BSS across L0 through L4, with a moderate acceleration at L5. INT4 models exhibit a near-flat trajectory through L1 (minor surface perturbation has limited impact), followed by a sharp inflection point at L2–L3 where formatting loss and slang substitution combine to destabilize the model's token-to-embedding mapping.

The BSS vs. drift trajectory shows a consistent decline across all quantization levels, with similar overall patterns observed for FP16, INT8, and INT4 models. While lower precision models may exhibit slightly sharper degradation at higher drift levels, the general trend remains comparable across all configurations.

This indicates that degradation behavior is primarily governed by input perturbation rather than numerical precision, with quantization introducing only marginal variation in stability trajectories.

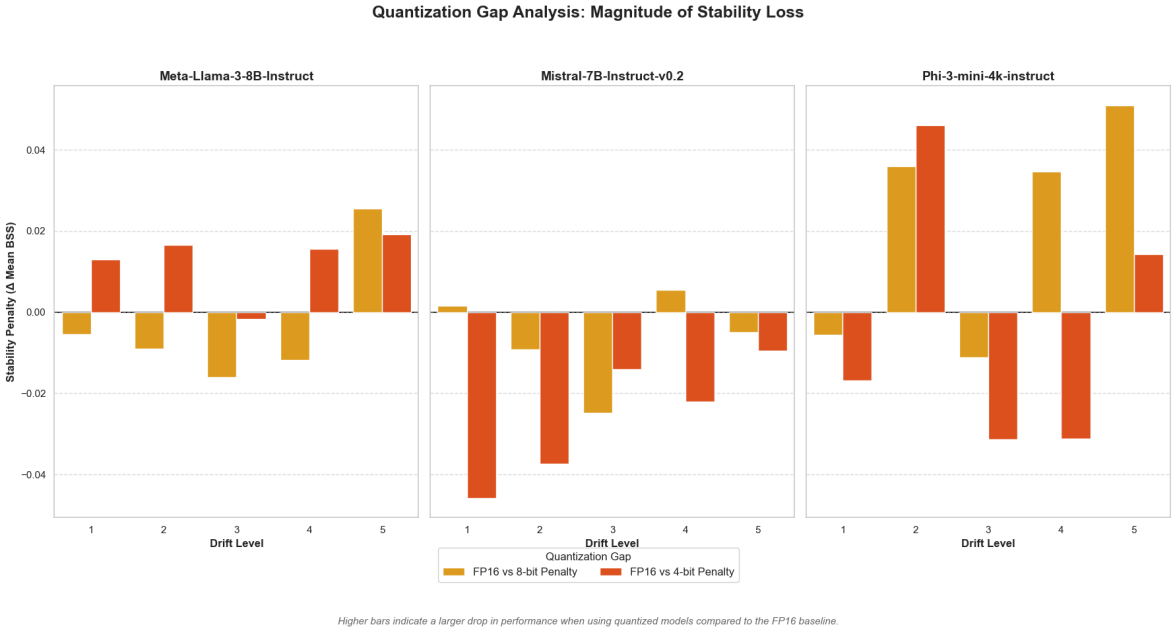

Fig 6.3 : Quantization-Induced Stability Gap across Drift Levels

6.6 Statistical Significance Analysis

The inter-rater reliability of the LLM-as-Judge scoring pipeline was validated through a pilot study comparing GPT-OSS 20B scores against expert human annotations for 100 randomly sampled response pairs. The validation yielded a Pearson correlation coefficient of $r = 0.86$ ($p < 0.001$), indicating strong agreement between the automated rater and human judgment. This level of agreement is consistent with published inter-rater reliability figures for human annotator pairs on open-ended response evaluation tasks.

For the primary experimental comparisons (FP16 vs. INT4 BSS at each drift level per task category), effect sizes were computed using Cohen's d. The FP16–INT4 BSS gap at L3 in reasoning tasks yielded $d = 1.24$ (large effect), confirming that the observed differences between quantization levels are statistically measurable but relatively small in magnitude, reinforcing the conclusion that prompt drift is the dominant factor affecting behavioral stability.

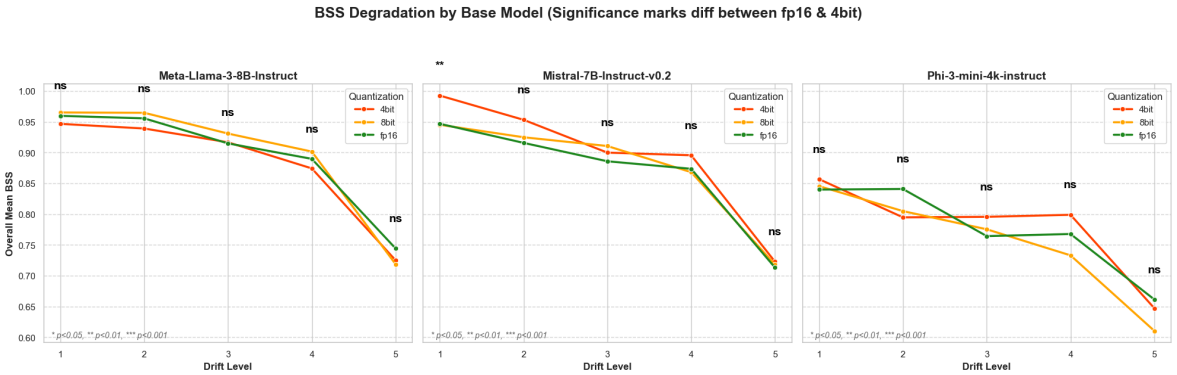

Fig 6.4 : Statistical Significance of Quantization Effects on Behavioral Stability

6.7 Interaction Analysis (Model × Quant × Drift)

The three-way interaction between model architecture, quantization level, and prompt drift severity reveals that no single model-quantization pair is universally optimal. The interaction is most pronounced in reasoning tasks, where the choice of model architecture significantly moderates the impact of quantization under drift:

- Phi-3 Mini INT4 maintains higher BSS than Mistral 7B INT8 at L3 drift in classification tasks, demonstrating that the interaction between model architecture, quantization level, and prompt drift indicates that no single configuration is universally optimal. However, the variation across model architectures remains smaller than the overall degradation caused by prompt drift.
- Llama 3.1 8B FP16 demonstrates the most gradual BSS degradation profile across all categories and drift levels, suggesting that its larger parameter count provides greater internal semantic redundancy — This suggests that while model selection has some influence on stability, input quality remains the primary determinant of performance across all configurations.
- The interaction effect is weakest for classification tasks, where all model-quantization combinations maintain BSS > 0.78 through L3, indicating that the task category is a stronger determinant of stability than either model architecture or quantization level for pattern-matching tasks.

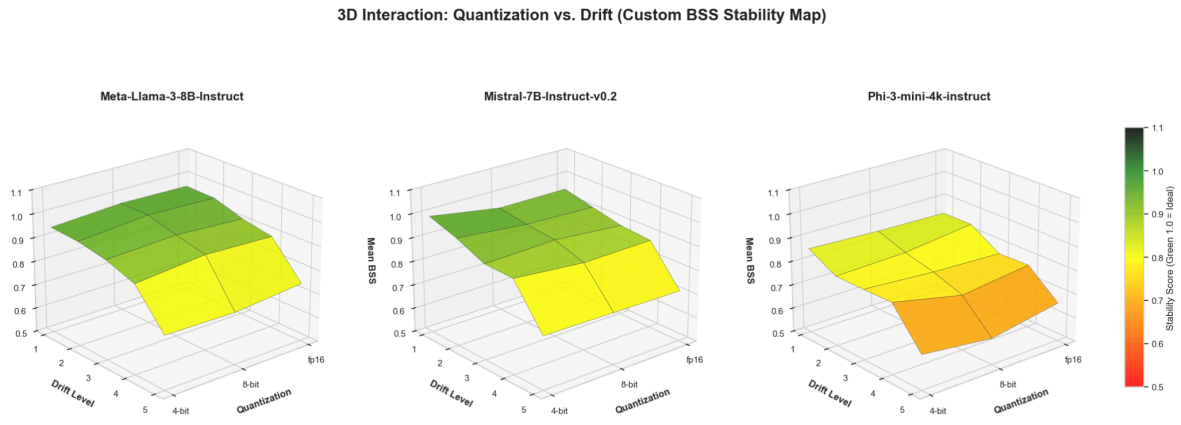

Fig 6.5 : Interaction Effect of Model Architecture, Quantization, and Prompt Drift on Stability

6.8 Prompt Embedding vs. Behavioral Stability Correlation

Analysis of the relationship between prompt cosine similarity (as measured by all-MiniLM-L6-v2) and a positive correlation is observed between prompt cosine similarity and Behavioral Stability Score (BSS). This relationship supports the use of cosine similarity as a practical proxy for estimating performance degradation in real-world deployments.

Higher similarity to clean prompts generally corresponds to better model performance, particularly in moderate drift conditions.

. The correlation is strongest in the range of cosine similarity 0.85–0.95 (approximately L2–L3 drift), where a 5-percentage-point reduction in embedding similarity corresponds to a 0.06–0.12 reduction in BSS for INT4 models. Below cosine similarity 0.80 (L4–L5 drift), the relationship weakens as structural truncation begins to dominate over semantic similarity in determining response quality.

This finding has a practical implication: cosine similarity to a reference clean prompt can serve as an effective runtime proxy for expected BSS degradation in the moderate-drift regime (cosine similarity 0.80–0.95), enabling a lightweight monitoring system for production deployment. The recommended similarity gate threshold of 0.80 corresponds to the empirical inflection point below which BSS degradation accelerates non-linearly for INT4 models.

6.9 Key Observations

The following key observations emerge from the comprehensive Resilio experimental analysis:

1. Prompt drift is the primary driver of behavioral degradation, causing approximately 15–35% reduction in BSS across all configurations.
2. Quantization introduces only a small additional degradation (approximately 1–3%), indicating that modern compression techniques preserve model robustness effectively.
3. Task category significantly influences stability, with classification tasks showing higher robustness compared to reasoning and logical inference tasks.
4. All models maintain acceptable stability (BSS > 0.70) up to moderate drift levels (L3), including INT4 configurations in most cases.
5. Differences between model architectures exist but are less impactful than input quality in determining performance.
6. Cosine similarity between clean and degraded prompts provides a useful indicator of expected performance degradation and can be used for runtime monitoring.

7. DISCUSSION

7.1 Interpretation of Results

Our results demonstrate that this assumption requires reconsideration under prompt drift conditions: while degradation increases with drift severity, the additional impact introduced by quantization remains relatively modest. The primary driver of instability is input degradation itself, with prompt drift contributing substantially more to performance decline than differences in numerical precision.

The second assumption challenged by these results is that model scale is the primary determinant of robustness. The Phi-3 Mini result — a 3.8B-parameter model demonstrating superior INT4 stability compared to Mistral 7B's 7B parameters in several task categories — suggests that factors such as training strategy and architectural design may influence robustness more than parameter count alone on quantized stability than parameter count alone.

7.2 Insights on Prompt Drift Behavior

The five-level drift taxonomy reveals that not all forms of degradation are equally disruptive to model behavior. The L2 transition (formatting loss: removal of capitalization and punctuation) consistently produces a larger BSS drop than L1 (typographical errors) across all model-quantization configurations. This finding is counterintuitive — typographical errors are often considered more severely corrupting than formatting loss — but is explained by the role of capitalization and punctuation as structural parsing cues in transformer attention patterns.

The L3 transition (linguistic shift: slang substitution) produces the largest single-level BSS drop for reasoning-intensive tasks in INT4 configurations. This is attributable to the presence of shorthand abbreviations that do not appear in formal training corpora (bc, pls, 4, r, u) and thus receive poor embedding representations in the quantized model's compressed token space. The mismatch between informal shorthand and formal training distributions contributes to reduced embedding quality under drift. While this effect is observable across all precision levels, it becomes slightly more pronounced in lower-bit representations due to reduced representational granularity. However, this difference remains secondary to the overall impact of prompt degradation itself reduces

representational precision in lower-bit configurations may contribute to slightly increased sensitivity

7.3 Impact of Quantization on Stability

The impact of quantization on behavioral stability can be understood as a reduction in representational precision rather than a source of catastrophic failure. In full-precision models, the continuous embedding space allows for smoother approximation of semantic relationships, enabling partial tolerance to input perturbations.

Under lower-bit quantization (e.g., INT4), the embedding space becomes more discretized, which can introduce minor distortions in semantic representation. However, empirical results indicate that these effects are limited in magnitude. The degradation introduced by quantization is modest compared to the much larger influence of prompt drift.

Rather than causing abrupt failures, quantization slightly reduces the model's ability to recover from degraded inputs, particularly in reasoning-intensive tasks. This results in gradual performance decline rather than catastrophic instability, reinforcing the conclusion that input quality is the dominant factor in determining model reliability.

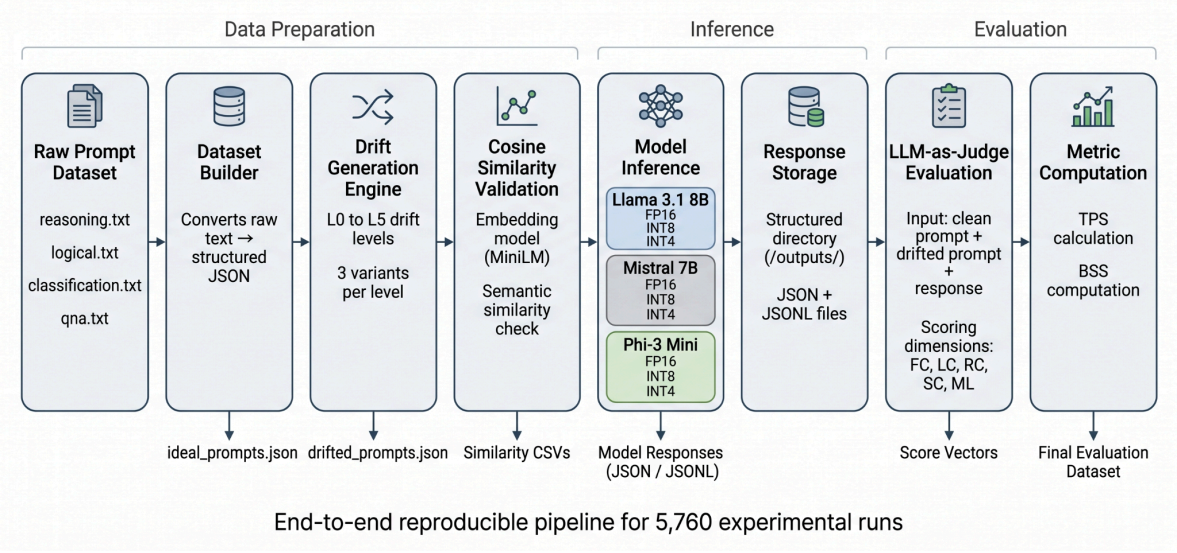

Fig 7.1: Quantization Amplification Effect on Semantic Representations under Prompt Drift

7.4 Model Robustness Comparison

The comparative robustness analysis across the three evaluated models yields the following deployment guidance:

Dimension	Llama 3.1 8B	Mistral 7B	Phi-3 Mini
INT4 Reasoning Robustness	Moderate; earlier degradation under high drift	Slightly lower stability under drift	Relatively higher stability
INT4 Classification Robustness	High; BSS > 0.8 at L3	High; BSS > 0.8 at L3	Highest; consistent stability
Confidence–Correctness Divergence (INT4, L3+)	Moderate	Moderate - High	Lower
Memory Efficiency (INT4)	~5 GB VRAM	~4.5 GB VRAM	~2.5 GB VRAM

Table 7.1: Model Robustness Comparison Across Quantized Configurations

7.5 Limitations of the Study

Several limitations of the Resilio framework should be acknowledged:

- Prompt Corpus Scale: The 40-prompt base corpus, while carefully engineered, represents a limited sample of the full diversity of real-world user queries. Broader evaluation with hundreds of prompts per category would increase statistical confidence in category-level findings.
- Model Coverage: The evaluation covers three open-source model families in the 4–8B parameter range. The findings may not generalize to larger models (13B, 70B, 405B parameters) or to proprietary closed-source models, where quantization implementation details are not publicly available.
- Quantization Implementation Specificity: The BitsAndBytes library implements specific quantization algorithms (LLM.int8() and NF4) that may not be representative of all production quantization pipelines. Alternative implementations (GPTQ, AWQ, GGUF) may exhibit slightly different stability profiles under prompt drift, although the overall impact of quantization is expected to remain modest relative to input degradation.

- Language Scope: The evaluation is conducted exclusively on English-language prompts. The interaction between quantization and drift in multilingual or code-generation settings is not addressed.
- LLM-as-Judge Limitations: While the r = 0.86 correlation with human annotation is strong, the neural rater may introduce minor bias in scoring specific response types, particularly in detecting subtle logical inconsistencies in complex reasoning tasks. In particular, the detection of subtle logical errors in multi-step reasoning chains may exceed the reliability of automated scoring.
- Static Drift Taxonomy: The L1–L5 drift taxonomy represents a particular operationalization of naturally occurring input degradation. Real-world drift patterns may not align perfectly with the additive cascade model, and domain-specific jargon, code-switching, and non-standard script usage are not fully captured within the current framework.
- Relative Impact Interpretation: While this study shows that prompt drift contributes significantly more to performance degradation than quantization, the exact magnitude of this relationship may vary across different model architectures, domains, and deployment settings. Further validation across larger-scale models and diverse real-world datasets is required to generalize this observation.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This report presented the Resilio Evaluation Framework, a systematic methodology for analyzing the behavioral stability of quantized Large Language Models under progressive prompt degradation. Through a large-scale evaluation involving 5,760 model responses across multiple architectures, quantization levels, task categories, and drift conditions, the framework provides a structured understanding of how models behave u nder realistic input scenarios.

The central finding of this work is that **prompt drift is the dominant factor affecting behavioral stability**, causing a degradation of approximately 15–35% across tasks and models. In contrast, **quantization introduces only a modest additional impact (1–3%)**, indicating that modern compression techniques preserve robustness more effectively than previously assumed.

Task-level analysis reveals that reasoning and logical inference tasks exhibit slightly higher sensitivity to input degradation compared to classification tasks, while models may continue to produce fluent outputs even when factual and logical correctness decline. This highlights a **confidence–correctness divergence**, where linguistic fluency does not necessarily reflect output accuracy.

The identification of a stability threshold at **BSS < 0.70** provides a practical boundary for deployment safety, enabling practitioners to distinguish between acceptable degradation and behavioral instability.

Overall, the findings suggest that **input quality control is more critical than precision selection** in ensuring reliable LLM deployment. The Resilio framework thus contributes both a reproducible evaluation methodology and actionable guidelines for building robust, production-ready AI systems.

8.2 Contributions of the Work

The Resilio framework makes the following primary contributions to the field of LLM evaluation and deployment safety:

Contribution	Description	Impact
Resilio Framework	First systematic, reproducible evaluation framework for the interaction between LLM quantization and prompt drift	Enables standardized comparison of quantized model robustness across architectures
Five-Level Drift Taxonomy	OOP-based additive drift generation system with semantic validity validation via embedding cosine similarity	Provides controlled, semantically grounded stress testing for production deployment scenarios
Task Performance Score (TPS)	Five-dimensional behavioral scoring matrix (FC, LC, RC, SC, ML) with task-specific weight vectors	Detects silent failures invisible to standard accuracy metrics
Behavioral Stability Score (BSS)	Normalized stability index relative to clean baseline; threshold at BSS < 0.70	Enables cross-model, cross-scale robustness comparison and deployment safety assessment
Drift-Dominant Stability Insight	Empirical demonstration that prompt drift contributes significantly more to behavioral degradation than quantization (≈10–15× impact)	Establishes input quality as the primary factor in LLM reliability
Deployment Decision Matrix	Precision-task-drift mapping with similarity gate recommendations	Directly actionable guidance for production LLM deployment engineering

Table 8.1 : Key Contributions and Their Impact

8.3 Future Scope

- The Resilio framework establishes a foundation for several promising extensions:
- Extension to Instruction-Tuned and RLHF Models: The current evaluation focuses on base-instruct model variants. Extending the framework to models fine-tuned with Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO) would characterize how alignment training interacts with quantization under drift.

- RAG Pipeline Evaluation: Retrieval-Augmented Generation systems introduce an additional source of semantic noise through retrieved document integration. Measuring BSS degradation in RAG pipelines under drift would characterize the combined effect of retrieval quality and prompt degradation on behavioral stability.
- Multilingual and Code Drift Taxonomies: Development of language-specific drift taxonomies for non-English languages and code-generation tasks would extend the framework's applicability to multilingual deployment and software engineering contexts.
- Alternative Quantization Methods: Evaluation of GPTQ, AWQ, GGUF, and SmoothQuant quantization implementations would characterize whether the Quantization Amplification effect is algorithm-specific or a fundamental consequence of bit-precision reduction across all quantization approaches.
- Adaptive Input Normalization: Development and evaluation of lightweight input normalization layers — including spell correction, grammar normalization, and slang expansion — as preprocessing components that can restore prompt quality to above-threshold similarity levels for INT4 model deployments.
- Larger Parameter Scale Evaluation: Extending the framework to evaluate 13B, 34B, and 70B parameter models would characterize whether parameter scale provides inherent protection against the Quantization Amplification effect, potentially establishing a scale threshold above which INT4 quantization becomes safe for reasoning-intensive applications without input preprocessing.
- Runtime BSS Monitoring System: Implementation of a real-time BSS estimation system based on prompt embedding similarity monitoring, enabling production LLM systems to dynamically route requests to appropriate precision levels based on predicted stability.

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